

Practical Laboratory 2

OPNsense Firewall Policy Testing Logging and Packet Analysis

Student identification

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The purpose of this lab is to configure and test firewall rules without losing general connectivity.

Part A: Record the baseline

To begin, commands were run to establish a baseline of operations for the Ubuntu client.

```
icdfa@icdfa-nsrab-client-v1:~$ ip -4 -br address
lo                UNKNOWN          127.0.0.1/8
ens33             UP                10.10.10.54/24
icdfa@icdfa-nsrab-client-v1:~$ ip route
default via 10.10.10.1 dev ens33 proto dhcp src 10.10.10.54 metric 100
10.10.10.0/24 dev ens33 proto kernel scope link src 10.10.10.54 metric 100
icdfa@icdfa-nsrab-client-v1:~$ ping -c 5 10.10.10.1
PING 10.10.10.1 (10.10.10.1) 56(84) bytes of data.
64 bytes from 10.10.10.1: icmp_seq=1 ttl=64 time=2.10 ms
64 bytes from 10.10.10.1: icmp_seq=2 ttl=64 time=0.519 ms
64 bytes from 10.10.10.1: icmp_seq=3 ttl=64 time=0.883 ms
64 bytes from 10.10.10.1: icmp_seq=4 ttl=64 time=1.01 ms
64 bytes from 10.10.10.1: icmp_seq=5 ttl=64 time=0.885 ms

--- 10.10.10.1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4056ms
rtt min/avg/max/mdev = 0.519/1.080/2.100/0.535 ms
icdfa@icdfa-nsrab-client-v1:~$ ping -c 5 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data.
64 bytes from 1.1.1.1: icmp_seq=1 ttl=127 time=20.5 ms
64 bytes from 1.1.1.1: icmp_seq=2 ttl=127 time=9.59 ms
64 bytes from 1.1.1.1: icmp_seq=3 ttl=127 time=39.6 ms
64 bytes from 1.1.1.1: icmp_seq=4 ttl=127 time=21.1 ms
64 bytes from 1.1.1.1: icmp_seq=5 ttl=127 time=32.2 ms

--- 1.1.1.1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4008ms
rtt min/avg/max/mdev = 9.586/24.591/39.553/10.350 ms
```

Establishing a baseline

```

icdfa@icdfa-ns-lab-client-v1:~$ getent hosts example.com
2606:4700:10::6814:179a example.com
2606:4700:10::ac42:93f3 example.com
icdfa@icdfa-ns-lab-client-v1:~$ curl -I https://example.com
HTTP/2 200
date: Tue, 06 Oct 2026 13:56:25 GMT
content-type: text/html; charset=utf-8
server: cloudflare
last-modified: Sun, 04 Oct 2026 20:44:03 GMT
allow: GET, HEAD
accept-ranges: bytes
age: 6191
cf-cache-status: HIT
cf-ray: a4652b5e3d2b9330-ACC
alt-svc: h3=":443"; ma=86400

icdfa@icdfa-ns-lab-client-v1:~$ curl -I http://example.com
HTTP/1.1 200 OK
Date: Tue, 06 Oct 2026 13:56:32 GMT
Content-Type: text/html; charset=utf-8
Connection: keep-alive
Server: cloudflare
Last-Modified: Fri, 02 Oct 2026 16:11:02 GMT
Allow: GET, HEAD
Accept-Ranges: bytes
Age: 6217
cf-cache-status: HIT
CF-RAY: a4652b87c91f932a-ACC
alt-svc: h3=":443"; ma=86400

```

Establishing that both HTTP and HTTPS work before restrictions are introduced

Block ICMP to one test address

Firewall: Rules

All rules Categories Search 500

	Icons	Version	Protocol	Source	Port	Destination	Port	Gateway	Description	Commands
Automatically generated rules										
Interface rules										
☐		LAN	IPv4	ICMP	LAN network	*	1.1.1.1/32	*	LAB2 BLOCK ICMP TO 1.1.1.1	
☐		LAN	IPv4	*	LAN network	*	*	*	Default allow LAN to any rule	
☐		LAN	IPv6	*	LAN network	*	*	*	Default allow LAN IPv6 to any rule	

Showing 1 to 31 of 31 entries

Apply

First firewall rule to block IPv4 ICMP traffic to 1.1.1.1

```

icdfa@icdfa-ns-lab-client-v1:~$ ping -c 4 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data.

--- 1.1.1.1 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3091ms

icdfa@icdfa-ns-lab-client-v1:~$ getent hosts example.com
2606:4700:10::6814:179a example.com
2606:4700:10::ac42:93f3 example.com
icdfa@icdfa-ns-lab-client-v1:~$ curl -I https://example.com
HTTP/2 200
date: Tue, 06 Oct 2026 15:40:41 GMT
content-type: text/html; charset=utf-8
server: cloudflare
last-modified: Fri, 02 Oct 2026 16:11:02 GMT
allow: GET, HEAD
accept-ranges: bytes
age: 6672
cf-cache-status: HIT
cf-ray: a465c419bb2cef19-LHR
alt-svc: h3=":443"; ma=86400

```

Tests to prove that ICMP traffic to 1.1.1.1 is blocked and DNS and HTTPS still work.

Block outbound HTTP while allowing HTTPS

Firewall: Rules

	Icons	Version	Protocol	Source	Port	Destination	Port	Gateway	Description	Commands
Automatically generated rules										
Interface rules										
☐		LAN	IPv4	ICMP	LAN network	*	1.1.1.1/32	*	LAB2 BLOCK ICMP TO 1.1.1.1	
☐		LAN	IPv4	TCP	LAN network	*	*	http	LAB2 BLOCK OUTBOUND HTTP	
☐		LAN	IPv4	*	LAN network	*	*	*	Default allow LAN to any rule	
☐		LAN	IPv6	*	LAN network	*	*	*	Default allow LAN IPv6 to any rule	

Showing 1 to 32 of 32 entries

Second firewall rule to block unencrypted web traffic (HTTP) on TCP port 80.

```
icdfa@icdfa-nslab-client-v1:~$ curl --max-time 10 -I http://example.com
curl: (28) Connection timed out after 10003 milliseconds
icdfa@icdfa-nslab-client-v1:~$
icdfa@icdfa-nslab-client-v1:~$ curl --max-time 10 -I https://example.com
HTTP/2 200
date: Tue, 06 Oct 2026 15:55:00 GMT
content-type: text/html; charset=utf-8
server: cloudflare
last-modified: Sun, 04 Oct 2026 20:44:03 GMT
allow: GET, HEAD
accept-ranges: bytes
age: 13306
cf-cache-status: HIT
cf-ray: a465d9125b49932e-ACC
alt-svc: h3=":443"; ma=86400
```

Test to prove that unencrypted web traffic (HTTP) is blocked and that encrypted web traffic (HTTPS) is not affected.

Examine firewall logs

Firewall: Log Files: Live View

Filters

Quick search (all fields)...

dst

is

1.1.1.1

Apply

Active filters:

dst is "1.1.1.1" ✕

Templates

>>

Choose template

✕

Options

☒ Auto-refresh
 ☐ Lookup hostname
 ☐ Select any of given

25

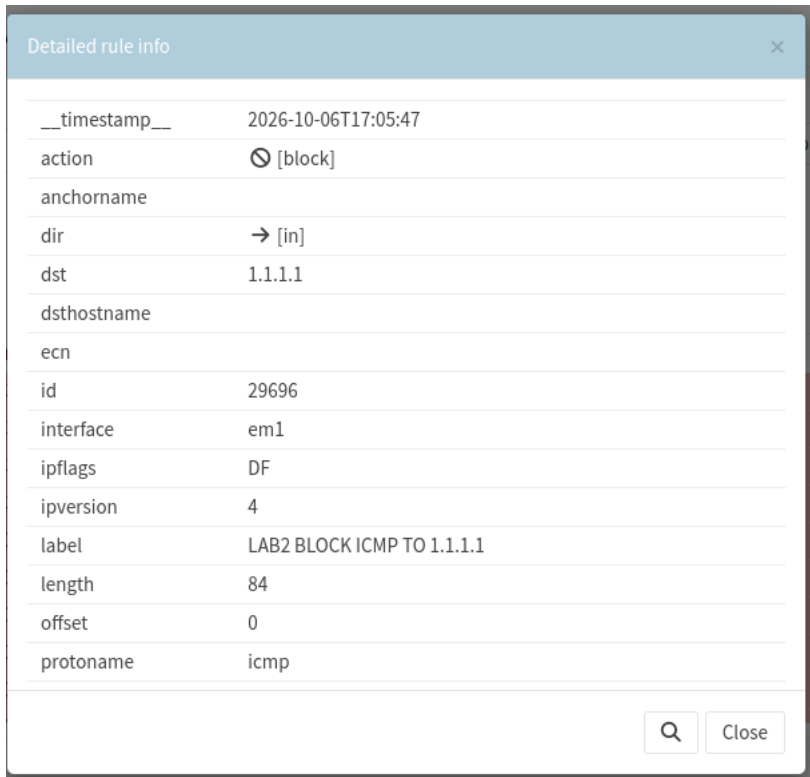
Table si

10000

History

Interface		Time	Protocol	Source	Destination	Action	Label
LAN	In	2026-10-06T17:05:47	ICMP	10.10.10.54	1.1.1.1	block	LAB2 BLOCK ICMP TO 1.1.1.1
LAN	In	2026-10-06T17:05:46	ICMP	10.10.10.54	1.1.1.1	block	LAB2 BLOCK ICMP TO 1.1.1.1
LAN	In	2026-10-06T17:05:45	ICMP	10.10.10.54	1.1.1.1	block	LAB2 BLOCK ICMP TO 1.1.1.1
LAN	In	2026-10-06T17:05:44	ICMP	10.10.10.54	1.1.1.1	block	LAB2 BLOCK ICMP TO 1.1.1.1
LAN	In	2026-10-06T17:03:09	ICMP	10.10.10.54	1.1.1.1	block	LAB2 BLOCK ICMP TO 1.1.1.1
LAN	In	2026-10-06T17:03:08	ICMP	10.10.10.54	1.1.1.1	block	LAB2 BLOCK ICMP TO 1.1.1.1
LAN	In	2026-10-06T17:03:07	ICMP	10.10.10.54	1.1.1.1	block	LAB2 BLOCK ICMP TO 1.1.1.1
LAN	In	2026-10-06T17:03:06	ICMP	10.10.10.54	1.1.1.1	block	LAB2 BLOCK ICMP TO 1.1.1.1
LAN	In	2026-10-06T17:02:06	ICMP	10.10.10.54	1.1.1.1	block	LAB2 BLOCK ICMP TO 1.1.1.1

Live View of Logs from the LAB2BLOCKICMPTO1.1.1.1 firewall rule



A detailed view of a log from the LAB2BLOCKICMPTO1.1.1.1 firewall rule.

Firewall: Log Files: Live View

Filters: Quick search (all fields)... [Refresh] [Reset]

dst_port is 80 [Apply]

Active filters: dstport is "80"

Templates: Choose template [Trash]

Options:

- ☒ Auto-refresh
- ☐ Lookup hostnames
- ☐ Select any of given criteria (or)

Table size: 25 | History size: 10000

Interface		Time	Protocol	Source	Destination	Action	Label	
LAN	In	2026-10-06T17:09:46	TCP	10.10.10.54:60080	151.101.65.91:80	block	LAB2 BLOCK OUTBOUND HTTP	🔍
LAN	In	2026-10-06T17:09:46	TCP	10.10.10.54:60074	151.101.65.91:80	block	LAB2 BLOCK OUTBOUND HTTP	🔍
LAN	In	2026-10-06T17:09:45	TCP	10.10.10.54:60080	151.101.65.91:80	block	LAB2 BLOCK OUTBOUND HTTP	🔍
LAN	In	2026-10-06T17:09:45	TCP	10.10.10.54:60074	151.101.65.91:80	block	LAB2 BLOCK OUTBOUND HTTP	🔍
LAN	In	2026-10-06T17:09:44	TCP	10.10.10.54:60080	151.101.65.91:80	block	LAB2 BLOCK OUTBOUND HTTP	🔍
LAN	In	2026-10-06T17:09:44	TCP	10.10.10.54:60074	151.101.65.91:80	block	LAB2 BLOCK OUTBOUND HTTP	🔍
LAN	In	2026-10-06T17:09:43	TCP	10.10.10.54:60080	151.101.65.91:80	block	LAB2 BLOCK OUTBOUND HTTP	🔍
LAN	In	2026-10-06T17:09:43	TCP	10.10.10.54:60074	151.101.65.91:80	block	LAB2 BLOCK OUTBOUND HTTP	🔍
LAN	In	2026-10-06T17:09:42	TCP	10.10.10.54:60080	151.101.65.91:80	block	LAB2 BLOCK OUTBOUND HTTP	🔍

Live view of Log files from the Firewall showing LAB2BLOCKOUTBOUNDHTTP rule

Detailed rule info	
__timestamp__	2026-10-06T17:09:45
ack	
action	[block]
anchorname	
datalen	0
dir	→ [in]
dst	151.101.65.91
dsthostname	
dstport	80
ecn	
id	10584
interface	em1
ipflags	DF
ipversion	4
label	LAB2 BLOCK OUTBOUND HTTP

The traffic matched the block rule instead of the default allow rule because the block rule was first on the list. OPNsense evaluates firewall policies from top to bottom.

Correlate with Wireshark

No.	Time	Source	Destination	Protocol	Length	Info
221	31.478934852	10.10.10.54	1.1.1.1	ICMP	98	Echo (ping) request id=0x27e5, seq=1/256, ttl=64 (no response found!)
228	32.502268319	10.10.10.54	1.1.1.1	ICMP	98	Echo (ping) request id=0x27e5, seq=2/512, ttl=64 (no response found!)
235	33.525943256	10.10.10.54	1.1.1.1	ICMP	98	Echo (ping) request id=0x27e5, seq=3/768, ttl=64 (no response found!)
246	34.549433534	10.10.10.54	1.1.1.1	ICMP	98	Echo (ping) request id=0x27e5, seq=4/1024, ttl=64 (no response found!)
589	92.601643368	10.10.10.54	1.1.1.1	ICMP	98	Echo (ping) request id=0x27ef, seq=1/256, ttl=64 (no response found!)
596	93.621875400	10.10.10.54	1.1.1.1	ICMP	98	Echo (ping) request id=0x27ef, seq=2/512, ttl=64 (no response found!)
609	94.646099756	10.10.10.54	1.1.1.1	ICMP	98	Echo (ping) request id=0x27ef, seq=3/768, ttl=64 (no response found!)
616	95.671018451	10.10.10.54	1.1.1.1	ICMP	98	Echo (ping) request id=0x27ef, seq=4/1024, ttl=64 (no response found!)

Filtered traffic to show the LAB2BLOCKICMPTO1.1.1.1 rule in effect (Protocol : ICMP and Destination Address: 1.1.1.1). Echo requests leave Ubuntu but no echo replies are observed.

Wireshark packet capture showing filtered traffic for `tcp.dstport == 80`. The capture shows multiple TCP SYN packets from 10.10.10.54 to 172.66.147.243 on port 80, but no corresponding SYN ACK packets are visible, indicating the firewall rule LAB2BLOCKOUTBOUNDHTTP is active.

No.	Time	Source	Destination	Protocol	Length	Info
129	20.022113785	10.10.10.54	104.20.23.154	TCP	74	[TCP Retransmission] 34712 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TS...
138	21.045451863	10.10.10.54	104.20.23.154	TCP	74	[TCP Retransmission] 34712 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TS...
147	22.069692438	10.10.10.54	104.20.23.154	TCP	74	[TCP Retransmission] 34712 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TS...
156	22.944349934	10.10.10.54	172.66.147.243	TCP	74	50992 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=141826180 TSecr=0...
163	23.989683199	10.10.10.54	172.66.147.243	TCP	74	[TCP Retransmission] 50992 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TS...
172	25.013511685	10.10.10.54	172.66.147.243	TCP	74	[TCP Retransmission] 50992 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TS...
181	26.037924952	10.10.10.54	172.66.147.243	TCP	74	[TCP Retransmission] 50992 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TS...
190	27.061744848	10.10.10.54	172.66.147.243	TCP	74	[TCP Retransmission] 50992 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TS...
729	116.579862067	10.10.10.54	172.66.147.243	TCP	74	57508 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2178662538 TSecr=...
736	117.621461471	10.10.10.54	172.66.147.243	TCP	74	[TCP Retransmission] 57508 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TS...
743	118.645629976	10.10.10.54	172.66.147.243	TCP	74	[TCP Retransmission] 57508 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TS...
752	119.669743972	10.10.10.54	172.66.147.243	TCP	74	[TCP Retransmission] 57508 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TS...
761	120.693524558	10.10.10.54	172.66.147.243	TCP	74	[TCP Retransmission] 57508 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TS...
770	121.582551967	10.10.10.54	104.20.23.154	TCP	74	35562 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2445221772 TSecr=...
777	122.613877830	10.10.10.54	104.20.23.154	TCP	74	[TCP Retransmission] 35562 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TS...
784	123.638623984	10.10.10.54	104.20.23.154	TCP	74	[TCP Retransmission] 35562 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TS...

Frame 190: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface
 Ethernet II, Src: VMware_03:b3:78 (00:0c:29:03:b3:78), Dst: VMware_0f:34:28 (00:0
 Internet Protocol Version 4, Src: 10.10.10.54, Dst: 172.66.147.243
 Transmission Control Protocol, Src Port: 50992, Dst Port: 80, Seq: 0, Len: 0

Filtered traffic to indicating that the firewall rule LAB2BLOCKOUTBOUNDHTTP is active. TCP SYN packets are transmitted but there are no SYN ACK packets to continue the handshake.

Wireshark packet capture showing filtered traffic for `tcp.port == 443`. The capture shows TLS and TCP ACK messages, indicating that HTTPS traffic is not affected by the rules set.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	10.10.10.54	10.10.10.1	TLSv1.2	188	Application Data
2	0.000419830	10.10.10.1	10.10.10.54	TCP	66	443 → 47490 [ACK] Seq=1 Ack=123 Win=514 Len=0 TSval=619933513 TSecr=1614623751
3	0.008133128	10.10.10.1	10.10.10.54	TLSv1.2	1013	Application Data
4	0.0088215168	10.10.10.54	10.10.10.1	TCP	66	47490 → 443 [ACK] Seq=123 Ack=948 Win=1157 Len=0 TSval=1614623839 TSecr=619933596
5	0.0088320758	10.10.10.1	10.10.10.54	TLSv1.2	97	Application Data
6	0.0088335718	10.10.10.54	10.10.10.1	TCP	66	47490 → 443 [ACK] Seq=123 Ack=979 Win=1157 Len=0 TSval=1614623839 TSecr=619933596
7	1.122046277	10.10.10.54	10.10.10.1	TLSv1.2	188	Application Data
8	1.122556566	10.10.10.1	10.10.10.54	TCP	66	443 → 47490 [ACK] Seq=979 Ack=245 Win=514 Len=0 TSval=619934635 TSecr=1614624873
9	1.209901975	10.10.10.1	10.10.10.54	TLSv1.2	1013	Application Data
10	1.209957755	10.10.10.54	10.10.10.1	TCP	66	47490 → 443 [ACK] Seq=245 Ack=1926 Win=1157 Len=0 TSval=1614624961 TSecr=619934716
11	1.210342124	10.10.10.1	10.10.10.54	TLSv1.2	97	Application Data
12	1.210367474	10.10.10.54	10.10.10.1	TCP	66	47490 → 443 [ACK] Seq=245 Ack=1957 Win=1157 Len=0 TSval=1614624962 TSecr=619934716
13	2.216590687	10.10.10.54	10.10.10.1	TLSv1.2	188	Application Data
14	2.217227994	10.10.10.1	10.10.10.54	TCP	66	443 → 47490 [ACK] Seq=1957 Ack=367 Win=514 Len=0 TSval=619935730 TSecr=1614625968
15	2.303528759	10.10.10.1	10.10.10.54	TLSv1.2	1013	Application Data
16	2.303595359	10.10.10.54	10.10.10.1	TCP	66	47490 → 443 [ACK] Seq=367 Ack=2904 Win=1157 Len=0 TSval=1614626055 TSecr=619935815

Frame 225: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface
 Ethernet II, Src: VMware_03:b3:78 (00:0c:29:03:b3:78), Dst: VMware_0f:34:28 (00:0
 Internet Protocol Version 4, Src: 10.10.10.54, Dst: 10.10.10.1
 Transmission Control Protocol, Src Port: 47490, Dst Port: 443, Seq: 3782, Ack: 43

This image indicates that HTTPS traffic is not affected by any of the rules set. TLS and TCP ACK messages are present.

Inspect states and automatic NAT

Firewall: Diagnostics: States

Select rule

States Actions

Search

50

	Int	Dir	Proto	Source	Nat	Destination	Gateway	State	Rule	Commands
☐	all	←	udp	192.168.219.165:46647		46.227.203.69:53		MULTIPLE:SINGLE	let out anything from fire...	
☐	all	←	udp	192.168.219.165:39729		45.127.113.23:53		MULTIPLE:SINGLE	let out anything from fire...	
☐	all	←	udp	192.168.219.165:123		162.159.200.1:123		MULTIPLE:SINGLE	let out anything from fire...	
☐	all	→	tcp	10.10.10.54:49090		172.66.147.243:443		FIN_WAIT_2:FIN_WAIT_2	Default allow LAN to any ...	
☐	all	←	tcp	192.168.219.165:12540	10.10.10.54:49090	172.66.147.243:443		FIN_WAIT_2:FIN_WAIT_2	let out anything from fire...	
☐	all	←	udp	192.168.219.165:24168		23.150.41.53:53		MULTIPLE:SINGLE	let out anything from fire...	
☐	all	←	udp	192.168.219.165:47281		185.209.85.151:53		MULTIPLE:SINGLE	let out anything from fire...	
☐	all	←	udp	192.168.219.165:58865		160.119.216.201:53		MULTIPLE:SINGLE	let out anything from fire...	
☐	all	→	tcp	10.10.10.54:40634		10.10.10.1:443		ESTABLISHED:ESTABLIS...	anti-lockout rule	

Showing 1 to 21 of 21 entries

State table showing LAN side- connection and its translated WAN-side state.

Firewall: NAT: Source NAT

Mode Automatic Source NAT rule generation

Categories

Search

500

	Version	Protocol	Source	Port	Destination	Port	Translate Sou...	Translate Sou...	Description	Co
Automatically generated rules										
L	WAN	IPv4	*	LAN network, Loopback network, DMZ network	*	500	WAN address	*	Auto created rule for ISAKMP	
L	WAN	IPv4	*	LAN network, Loopback network, DMZ network	*	*	WAN address	*	Auto created rule	

Showing 1 to 2 of 2

Image showing the default NAT settings

Restore the laboratory

Firewall: Rules

	Icons	Version	Protocol	Source	Port	Destination	Port	Gateway	Description	Commands	
Automatically generated rules											
Interface rules											
☐	L	☐	LAN	IPv4	ICMP	LAN network	*	1.1.1.1/32	*	LAB2 BLOCK ICMP TO 1.1.1.1	Edit Delete Search
☐	L	☐	LAN	IPv4	TCP	LAN network	*	*	http	LAB2 BLOCK OUTBOUND HTTP	Edit Delete Search
☐	L	☒	LAN	IPv4	*	LAN network	*	*	*	Default allow LAN to any rule	Edit Delete Search
☐	L	☒	LAN	IPv6	*	LAN network	*	*	*	Default allow LAN IPv6 to any rule	Edit Delete Search

Showing 1 to 32 of 32 entries

Disabling the firewall rules to return to baseline settings.

```
icdfa@icdfa-nslab-client-v1:~$ ping -c 4 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data:
64 bytes from 1.1.1.1: icmp_seq=1 ttl=127 time=8.78 ms
64 bytes from 1.1.1.1: icmp_seq=2 ttl=127 time=7.42 ms
64 bytes from 1.1.1.1: icmp_seq=3 ttl=127 time=6.69 ms
64 bytes from 1.1.1.1: icmp_seq=4 ttl=127 time=8.15 ms

--- 1.1.1.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 6.685/7.758/8.782/0.785 ms
```

Ping testing to ensure that internet connectivity

```
icdfa@icdfa-ns-lab-client-v1:~$ curl --max-time 10 -I http://example.com
HTTP/1.1 200 OK
Date: Tue, 06 Oct 2026 17:18:03 GMT
Content-Type: text/html; charset=utf-8
Connection: keep-alive
Server: cloudflare
Last-Modified: Fri, 02 Oct 2026 16:11:02 GMT
Allow: GET, HEAD
Accept-Ranges: bytes
Age: 3907
cf-cache-status: HIT
CF-RAY: a46652ba2d23033a-ACC
alt-svc: h3=":443"; ma=86400

icdfa@icdfa-ns-lab-client-v1:~$ curl --max-time 10 -I https://example.com
HTTP/2 200
date: Tue, 06 Oct 2026 17:18:10 GMT
content-type: text/html; charset=utf-8
server: cloudflare
last-modified: Fri, 02 Oct 2026 16:11:02 GMT
allow: GET, HEAD
accept-ranges: bytes
age: 3894
cf-cache-status: HIT
cf-ray: a46652e4a83d033b-ACC
alt-svc: h3=":443"; ma=86400
```

Tests to ensure that HTTP and HTTPS traffic is allowed.

Analysis questions

1. Why must the specific block rules be placed above the broad allow rule?
Block rules must be placed above the broad allow rules because OPNsense evaluates firewall rules from top to bottom. If the broad allow rules are first, every packet will match it and be allowed to pass, so the block rules below will never be reached.
2. Which five packet attributes are most useful when explaining a firewall decision?
Source IP address, Destination IP address, Protocol, Destination Port, Interface and direction
3. Why did blocking ICMP not block HTTPS?
Firewall rules are protocol and traffic specific. ICMP and HTTPS are different protocols hence blocking one does not affect the other.
4. What difference did you observe between the blocked TCP port 80 traffic and permitted TCP port 443 traffic?
Blocked TCP hangs for a while and then times out. The permitted TCP port shows the complete TCP Handshake and curl returns a response.
5. What role does outbound NAT play when icdfa-ns-lab-client-v1 uses a private IPv4 address?
Outbound NAT rewrites the source address to the firewall's WAN address for communication with devices outside of the local subnet. Replies come back to the WAN address and the state table maps them back to the private IP address that requested the data.

6. Why is restoring the original state an important part of a controlled security laboratory?
Restoring the original state makes it easier to continue to another lab from a known baseline. Also, it sanitizes the lab environment, making sure that there are no open vulnerabilities.